

Fix a set X , and define the free monoidal category generated by X .
 It consists of objects generated by unit I and elements of X over a binary operation.
 Morphisms are generated by uniterable associates and unitors, composition, identity, and parallel action over $\otimes$ then quotiented by associate/unitor/pentagon/triangle laws + category laws.

This forms a monoidal category by construction, and it has a universal property of having a global elim of any displayed monoidal category displayed over it. i.e. a global elim.

$\exists ! \int^{M^D} \text{displayed monoidal} \xrightarrow{\quad} \text{FreeMon}(X)$ *needs explanation*

Moreover, for monoidal categories M, N we can show that weakening construction displaying M over N has the structure of a displayed category over N .

In particular, viewing a monoidal category M over $\text{FreeMon}(X)$ allows us to use the global elimination principle to extract the data of a strong monoidal functor from $\text{FreeMon}(X)$ to M .

$\hookrightarrow$ Note, the details that all generators from X must be interpreted in the monoidal category to make the eliminator.

Call this strong monoidal functor $\text{rec} : \text{FreeMon}(X) \rightarrow M$, the recursor.

The recursor is unique among all strong monoidal functors from $\text{FreeMon}(X)$ to M along an interpretation of generators $\gamma : X \rightarrow M$.

Thus if $G : \text{FreeMon}(X) \rightarrow \text{Mon}(X)$ is strong monoidal such that

$G(I, x) \equiv x$ then we can build a retract of the recursor.

$\hookrightarrow$ Double section/retraction terminology

That is, we can find a functor G^{-1} such that $G \circ G^{-1} \equiv \text{Id}_{\text{FreeMon}(X)}$.

$\hookrightarrow$ Elaborate on Is Fiber and IsoFiberReflex

$\hookrightarrow$ Rough Idea, given cats C, D , $F : C \rightarrow D$ define Iso as a displayed category $\int^{D \times D}$

describing isomorphisms between elements in the base category.

We can view Iso along Id_X .

$$\begin{array}{ccc} \text{reindex Iso}_D & & \text{Iso}_D \\ \downarrow & \text{Id}_X & \downarrow \\ D \times C & \xrightarrow{\text{Id}_X F} & D \times D \end{array}$$

Say $\text{IsoFiber}_F := \text{reindex}_{\text{Id}_X F} \text{Iso}_D$

Then a section of IsoFiber_F contains all data of a retraction.

$\hookrightarrow$ check section & retraction terminology

$D \xrightarrow{G} C \xrightarrow{F} D$ If $F \circ G \equiv \text{Id}_D$ then G is a section of F , F is a retraction of G .

To see this, note that a global section s of IsoFiber_F is given by

$$\int^{D \times C} s$$

Given a pair (d, c) , s picks out an isomorphism between d and $F(c)$. This specifies the object c that d should be mapped to.

$$G(d) := c$$

and manifestly $F(G(d)) = F(c) \cong d$

so $F \circ G \equiv \text{Id}$ intuitively (ignoring how morphisms are handled).